

Software Outline — Senior Design

Software files

- `Current_Complete_Product.ino`: the integrated product. Runs the white strobe, green chaser, and tone sweep while the trigger is held, and prints mic readings over serial.
- `White-LED-Strobe.ino`: prototype of the white LED strobe patterns.
- `Green-LED-Strobe.ino`: prototype of the rotating, blinking green LED chaser.
- `Speaker_Setup.ino`: prototype of random speaker volume patterns, with pitch tied to volume.
- `Light_Log_Without_SD_W_Comments.ino`: runs the light routine and logs every LED change over serial.
- `Light_Log_With_SD.ino`: runs the light routine and logs every LED change to an SD card.
- `Sound_Log_Without_SD_Comments.ino`: logs microphone readings over serial.
- `Sound_Log_With_SD_Comments.ino`: logs microphone readings to an SD card.
- `Software.code-workspace`: VS Code workspace settings; not part of the firmware.

Current_Complete_Product.ino

Functions

- `setup()`: sets pins 2–13 as outputs and A0 (trigger), A1 (mic) as inputs; starts serial at 115200 baud.
- `loop()`: while A0 is HIGH, runs one of ten random strobe patterns, then `runGreen()` and `speaker()`. When A0 goes LOW, turns pins 2–13 off, stops the tone, resets the green state, and prints a mic reading.
- `flicker(thisPin, duration, brightness)`: strobes `thisPin` for `duration / 70` on/off cycles of `freqWhite` ms each, running `runGreen()` and `speaker()` during the waits. Returns early if A0 goes LOW.
- `nbDelay(ms)`: non-blocking pause; waits `ms` while running `runGreen()` and `speaker()`. Returns early if A0 goes LOW.
- `runGreen()`: every `rotSpeed` ms, moves the three-pin green window one pin along pins 3–12; every `freqGreen` ms, toggles the window on or off.
- `speaker()`: prints a mic reading (time, raw, dB); every sweep ms, plays `speakerFreq` on pin 13 and raises it 100 Hz, wrapping from 3000 back to 1000 Hz.
- `logEvent(type, pin, stateStr)`: prints a CSV line (time, type, pin, state) while A0 is HIGH. Currently not called.

Key variables

- `lowestPin`, `highestPin` (2, 13): range of output pins set up and cleared.
- `freqWhite` (40 ms): white strobe half-period, giving 12.5 Hz.
- `freqGreen` (35 ms): green blink half-period, about 14.3 Hz.
- `rotSpeed` (28 ms): time per one-pin step of the green window.
- `state` (0 or 1): whether the green window is lit.
- `startPin` (3–12): first pin of the three-pin green window.

- speakerFreq (1000–3000 Hz): current speaker tone.
- sweep (50 ms): time between tone steps.
- lastTime1, lastTime2, lastTime3 (ms): last update times for green rotation, green blink, and tone step.
- sensorValue (0–1023): raw microphone reading from A1.
- dB (30–134): mic reading rescaled as $30 + (\text{sensorValue} / 1023) \times 104$.

White-LED-Strobe.ino

Functions

- setup(): sets pins 2–13 as outputs and A0 (trigger), A1 (mic) as inputs.
- loop(): while A0 is HIGH, picks a random pattern with random(0, 9) and runs its three strobe bursts and pauses using delay().
- flicker(thisPin, duration, brightness): strobes thisPin for duration / 70 on/off cycles of freq ms each.

Key variables

- lowestPin, highestPin (2, 13): range of output pins set up.
- freq (40 ms): strobe half-period, computed as $1000 \div (2 \times \text{desired Hz})$.
- thisPin, duration, brightness: global copies of the flicker() parameter names; not used.

Green-LED-Strobe.ino

Functions

- setup(): sets pins 2–13 as outputs and A0 (trigger), A1 (mic) as inputs.
- loop(): while A0 is HIGH, moves the three-pin green window one pin every rotSpeed ms and toggles it on or off every freq ms.

Key variables

- lowestPin, highestPin (2, 13): range of output pins set up.
- freq (35 ms): green blink half-period.
- rotSpeed (28 ms): time per one-pin step of the window.
- state (0 or 1): whether the window is lit.
- startPin (3–12): first pin of the three-pin window.
- lastTime1, lastTime2 (ms): last update times for rotation and blink.
- thisPin, duration, brightness: carried over from the white prototype; not used.

Speaker_Setup.ino

Functions

- `setup()`: sets pins 13 (speaker) and 12 (volume) as outputs and A0 as input, writes the starting volume, and seeds `random()` from unconnected pin A3.
- `loop()`: while A0 is HIGH, starts a random volume pattern if none is active, then runs `volume()` and `speaker()`. When A0 goes LOW, stops the tone, sets volume to 0, and clears the pattern.
- `speaker()`: every sweep ms, maps `volumeLevel` (100–255) to a pitch of 3000–10000 Hz and plays it on pin 13.
- `volume()`: looks up the current step's level and duration in one of ten three-step patterns, limits the level to 100–255, and writes it as PWM on pin 12. After each step's duration it advances; after the third step it picks a new random pattern.

Key variables

- `speakerPin`, `volumePin`, `triggerPin` (13, 12, A0): speaker output, volume PWM output, trigger input.
- `minimumVolume`, `maximumVolume` (100, 255): limits for the volume level.
- `minimumFrequency`, `maximumFrequency` (3000, 10000 Hz): pitch range mapped from the volume range.
- `sweep` (50 ms): how often the pitch is updated.
- `volumeLevel` (100–255): current PWM volume level.
- `volumePattern` (–1 or 0–9): active pattern; –1 means none.
- `volumeStep` (0–2): current step within the pattern.
- `volumePatternStart` (ms): time the current step began.
- `lastTime3` (ms): time of the last pitch update.

Light_Log_Without_SD_W_Comments.ino

Functions

- `setup()`: sets pins 2–13 as outputs and A0, A1 as inputs; starts serial at 115200 baud.
- `loop()`: while A0 is HIGH, runs one of ten random strobe patterns, then `runGreen()`.
- `flicker(thisPin, duration, brightness)`: strobes `thisPin` as in the main product and logs each on and off.
- `nbDelay(ms)`: waits ms while running `runGreen()`.
- `runGreen()`: runs the green chaser as in the main product and logs each LED change.
- `logEvent(type, pin, stateStr)`: prints a CSV line: time, LED type, pin, ON or OFF.

Key variables

- `lowestPin`, `highestPin` (2, 13): range of output pins set up.
- `freqWhite`, `freqGreen`, `rotSpeed` (40, 35, 28 ms): white strobe half-period, green blink half-period, and green step time.
- `state` (0 or 1), `startPin` (3–12): whether the green window is lit, and its first pin.

- lastTime1, lastTime2 (ms): last update times for green rotation and blink.

Light_Log_With_SD.ino

Functions

- setup(): sets pins 2–13 as outputs and A0, A1 as inputs; starts serial at 9600 baud for error messages. Starts the SD card, opens strobe.txt, and writes the header row; halts if either fails.
- loop(): while A0 is HIGH, runs one of ten random strobe patterns, then runGreen(), and flushes the log file if at least 500 ms have passed.
- flicker(thisPin, duration, brightness): strobes thisPin as in the main product and writes each on and off to the file.
- nbDelay(ms): waits ms while running runGreen().
- runGreen(): runs the green chaser as in the main product and writes each LED change to the file.

Key variables

- chipSelect (10): SD card chip select pin.
- logFile: the open strobe.txt log file.
- lastFlush (ms): time the file was last flushed to the card.
- lowestPin, highestPin (2, 13): range of output pins set up.
- freqWhite, freqGreen, rotSpeed (40, 35, 28 ms): white strobe half-period, green blink half-period, and green step time.
- state (0 or 1), startPin (3–12): whether the green window is lit, and its first pin.
- lastTime1, lastTime2 (ms): last update times for green rotation and blink.

Sound_Log_Without_SD_Comments.ino

Functions

- setup(): starts serial at 115200 baud, sets A0 and A2 as inputs and pin 13 as output, and prints the header row Time(ms), Raw, dB.
- loop(): while A0 is HIGH, reads the mic, prints time, raw value, and dB, plays a 200 Hz tone on pin 13, and waits 200 ms. Stops the tone otherwise.

Key variables

- micPin (A2): microphone input.
- triggerPin (A0): trigger input; logging runs only while it is HIGH.
- chipSelect (10): SD chip select pin; declared but not used in this serial version.
- sensorValue (0–1023): raw microphone reading.
- dB (30–134): mic reading rescaled as $30 + (\text{sensorValue} / 1023) \times 104$.

Sound_Log_With_SD_Comments.ino

Functions

- `setup()`: starts serial at 9600 baud, sets A0 and A2 as inputs, and starts the SD card, halting if it fails. Deletes the old `log.txt` and writes the header row Time(ms), Raw, dB.
- `loop()`: while A0 is HIGH, reads the mic, opens `log.txt`, appends time, raw value, and dB, closes the file, and waits 200 ms.

Key variables

- `micPin` (A2): microphone input.
- `triggerPin` (A0): trigger input; logging runs only while it is HIGH.
- `chipSelect` (10): SD card chip select pin.
- `myFile`: the `log.txt` file, opened and closed for each sample.
- `sensorValue` (0–1023): raw microphone reading.
- `dB` (30–134): mic reading rescaled as $30 + (\text{sensorValue} / 1023) \times 104$.